

Aboriginal People and Groundwater

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Abstract

Aboriginal people have been part of the Australian landscape for 65,000 years or more, and in many areas, including the Great Artesian Basin, they have relied on groundwater for survival. Aboriginal people believe their story originated in the Dreamtime – the beginning, when Aboriginal cultural heroes created groundwater sites along with all other sacred sites. Their survival, particularly in a desert environment, has intrigued non-Aboriginal people for many years. While many studies have been conducted on how Aboriginal people survived at a local or regional level by accessing groundwater, no research has collated and reviewed the entire subject matter of ‘Aboriginal People and Groundwater’. This paper, based on my 2005 Masters Thesis, endeavours to collate and review available research and provide an insight into the cultural relationships and dependence of Aboriginal people on groundwater. Since colonisation, the Australian continent, its landscape and the complex nature of Aboriginal society have changed. So too have human uses and reliance on groundwater, for it has become a favoured water supply for many communities and types of industry. In some cases, these uses have led to over-allocation and groundwater depletion or degradation. The future of groundwater use has to be managed sustainably, as Aboriginal people have done for thousands of years.

Keywords: Indigenous cultural values, Dreamtime stories, rainbow serpent, springs, sustainable management of groundwater

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Introduction

Groundwater is an integral part of the total water cycle and is necessary to sustain human and ecological life. Essentially, it is all the water below the ground surface stored in aquifers. Its volume greatly exceeds all other freshwater sources that are unfrozen. Considering this, groundwater contributes to a large portion of water supplies in Australia, where reliance on surface water may not be a viable option in a dry landscape. For instance, in Perth, Western Australia, groundwater contributes to approximately 40% of the city’s water supply (Australian Water Association, 2005).

The occurrence of groundwater is dependent on local and/or regional geology, which can be complex. In Australia, aquifers can be classified into two types: confined aquifers and unconfined aquifers. Groundwater moves slowly through these aquifers –

usually less than one metre per day – until it seeps into low-lying areas, streams, lakes, wetlands or the ocean. It can also be forced out due to gravity or hydrostatic pressure, to discharge along geologic fault lines. Unlike surface water, groundwater losses through evaporation are low and are buffered against climatic variability, especially drought. Groundwater is a vast resource with relatively constant chemistry and temperature.

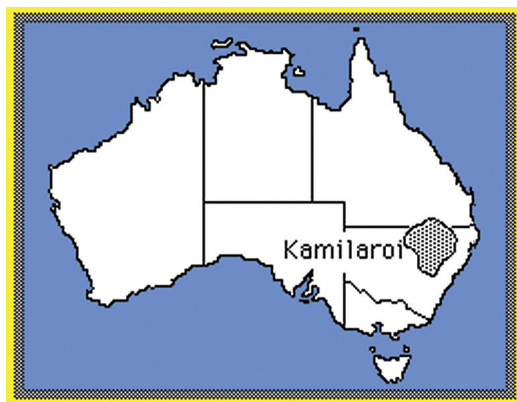
Because groundwater has become a favoured water supply option for many population centres around Australia, many aquifers are over-allocated and authorities now need to consider sustainable groundwater extraction and use. This is the case for many of the New South Wales (NSW) major aquifers, with over-allocation as an outcome of past government policies (Gates & O’Keefe, 2002). Pollution also threatens the quality of groundwater.

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Aboriginal communities are a part of the population centres that access groundwater for potable water. However, prior to European colonisation, Aboriginal people would have used groundwater sustainably and principally for survival; for them it was an infinite resource for thousands of years. Aboriginal water use is described in Lloyd (1988): “Australian Aborigines were exploiters, conservators, managers and manipulators of water resources. They were able to prevent the pollution of water, to filter it before drinking, to reticulate it, and to store it to reduce evaporation. Indeed, very little of the fundamental elements of hydrology and hydraulics eluded them.”

At the time of my Masters Thesis (Moggridge, 2005), there had been limited localised, regional or state-wide studies on Aboriginal people and groundwater – a national review was required to fill this knowledge gap. To this end, all available written or recorded research, personal accounts, audiovisual materials, reports, journals, conference papers, art forms, oral histories and interviews with elders were accessed to document the relationships of Aboriginal people with groundwater. This paper covers groundwater stories from the Dreamtime, through art and oral history, early observations of Aboriginal people and how they obtained water, and finally how Aboriginal people use groundwater today. The topic was more than appealing to me as a Masters research project, being of Aboriginal heritage from the Kamilaroi nation in north-western NSW (Figure 1).

Figure 1. Location of the Kamilaroi nation (Source: Austin & Nathan, 1998). Available at <https://www.dnathan.com/language/gamilaraay/dictionary/>



Moreover, my country is situated at the lower limits of the Great Artesian Basin that was so important to my ancestors and is necessary for the ongoing survival of my nation today.

Aboriginal People in the Australian Landscape

Aboriginal people have been part of Australia's landscape for millennia. There is no exact date of when Aboriginal people first arrived on the Australian continent or satisfactory evidence to indicate they evolved in Australia, but estimates range from 40,000 to 65,000 years. Considering this, many Aboriginal nations believe that they have been here since time began – since the Dreamtime “when the ancestral heroes first appeared and began their epic journey across the land” (Isaacs, 1984), and began creating the land, sky, water and life.

Since Aboriginal people have been a part of Australia's landscape, they have experienced and effected numerous changes to its environment: for instance, changes in flora and fauna, ecological communities, fire dependence, volcanic activity, climate, and water availability. The biggest change to the Aboriginal environment would have to be the invasion and settlement of Europeans and consequent rapid deterioration of their homelands.

Throughout all of these changes, Aboriginal people have adapted and survived. Their survival would not be possible without the knowledge of how to find and manage water – without water there is no life. Groundwater sources played a significant role in their survival, especially in the desert regions, which cover approximately 70% of the continent.

Aboriginal Groundwater Resources

Several groundwater-related water sources that Aboriginal people have used in the past, and still use today, will be described in this paper. However, to each Aboriginal tribe, their names and significance would differ. These resources include: soaks/soakages or native wells, springs, mound springs, bores and hanging swamps.

Soaks/soakage or native wells refer to places where groundwater discharges to the ground surface. Landform and vegetation are key indicators. They can occur near rivers, in ephemeral riverbeds and sand hills, and near salt lakes. Native wells are

simply the traditional means of tapping these soaks (Kavanagh, 1984), and were thus dug in areas where soakages were known. According to Bayly's study, they ranged in depth up to 7 metres but averaged 1.5 metres in depth; they were then filled with debris, sticks, sand, or had covers/caps placed on top of the wells to reduce evaporation and stop animals accessing them and fouling the water (Bayly, 1997). Bandler (2003) also mentions that some were curved in shape for protection from evaporation and were larger at the base to give greater capacity. Hercus & Clarke (1986) give a detailed description of nine wells in the Simpson Desert.

Springs are often confused with soakages, but Tweedie (cited in Kavanagh, 1984) explains that springs occur where impervious layers outcrop; the water table may be forced to the surface and water appears as a spring. Either gravitational or hydrostatic pressures force spring water to the surface.

Mound springs are geomorphic formations raised above the surrounding land surface, formed by a deposit of minerals and sediment brought up from the artesian aquifers or confining beds by water at certain natural discharge points (Great Artesian Basin Consultative Council, 2000).

Bores are structures drilled or dug below the surface to obtain water from an aquifer system (Murray-Darling Basin Commission, 1999).

Hanging swamps are another groundwater-dependent system that Aboriginal people would have used. Hanging swamps are shallow depressions on sloping rock faces on the edge of predominantly sandstone cliffs and occur at moderate to high altitudes. They are a constant source of water, fed by rain and groundwater, thus allowing a range of plant species which, combined, attract animal species.

Aboriginal People and Mound Springs

Studies on Aboriginal people and mound springs have involved archaeologists such as Lampert (1985) and Florek (1987), social scientists, mining enterprises (e.g. the Olympic Dam Project; SADEP, 1986) and governments, because springs hold great cultural and ecological significance.

Many stories of Aboriginal ancestral heroes are associated with mound springs and their placement along travel routes, and feature myths and spiritual significance. Mound springs are semi-permanent oases in the desert that have provided water for

Aboriginal people for thousands of years and thus have a strong cultural significance. All individual springs and complexes are known to hold significance for Aboriginal people, and it is impossible in contemporary times to predict, with any confidence, that an individual mound spring does not have any particular significance due to similarities with other springs in an area (Noble et al., 1998, in Mudd, 1998). Hercus & Sutton (1985) emphasise that "the springs are considered so important that the large-scale deterioration of any group of springs would cause great distress to at least some Aboriginal people, whether their associations with the sites are direct or indirect".

A paper by Ah Chee (2002) discusses the significance and deterioration of the Dalhousie Springs to the Indigenous Southern Aranda people and the Irrwanyere Aboriginal Corporation. The Dalhousie Springs are situated on the edge of the Simpson Desert in northern South Australia. To the Aranda the springs are known as the Irrwanyere or 'the healing springs'. Following European settlement, the springs became 'sick' from poor land management practices by settlers, and now the local Aboriginal people are left with a legacy of degraded land. It must be so painful for the traditional owners of Dalhousie Springs and also for all other traditional owners to see their land destroyed and treated with little or no respect; and when it is destroyed, the landholder sells it or leaves it for the Aboriginal people to repair. McFarlane (2004) explains the difficulty non-Aboriginal people may have in understanding Aboriginal people's distress when the environment is polluted or damaged: "If one part is damaged or destroyed, all other parts are under pressure."

Groundwater and the Dreamtime

Storytelling is an integral part of life for Australian Aboriginal peoples. Stories are passed from one generation to another, usually by the elders of Aboriginal communities. The Dreaming or Dreamtime is an English translation of an Aboriginal concept. For example, three tribes of central Australia, the Pitjantjatjara, Arrernte and Adnyamathanha, use the terms 'Tjukurpa', 'Aldjerinya' and Nguthuna', respectively, whereas the Gamilaraay people refer to the Dreamtime as 'Burruguu'.

Aboriginal stories are told in detail and re-enacted in ceremonies that capture the imagination

of the young, primarily for education. These teaching styles have proven to be inspiring and powerful tools in presenting Dreamtime beliefs and cultural practices. Dreamtime stories depict the very basic part of a long and complex event. Stories covered include: the creation of the land and life, protocols and tribal lore, life and death, warfare, hunting, linking every creature and every feature of the landscape, male and female roles, as well as sacred and public affairs.

These are stories of the history and culture of Aboriginal people, handed down in this way since the beginning of time; they refer to all that is known and all that is understood. Many groundwater-related sites would be Dreaming sites because water that originates from below the ground would be deemed spiritually significant by Aboriginal people. The Dreaming significance of these sites, for instance, would link surface and sub-surface waters through cultural heroes. These sites would have been the focus of trade, dispute resolution between tribes, male initiation and marriage. To Aboriginal people, the stories of the Dreamtime represent the past, present and future.

The Rainbow Serpent

The Rainbow Serpent was a highly significant and powerful cultural hero of Australian Aboriginal Dreamtime. The author has a cultural connection to the Kamilaroi's cultural hero, *Gurria* or *Kurrea*.

The Rainbow Serpent was connected to many different Aboriginal tribes throughout Australia, had many different names, and usually took the form of a snake/serpent creature that linked the people to features in the landscape. Sites and stories associated with the Rainbow Serpent were, and still are, considered culturally significant to Aboriginal people.

Anthropologists have long appreciated the significance of the Rainbow Serpent. Radcliffe-Brown (1930) stated that the Rainbow Serpent story was perhaps "the most important of the mythology and that fuller knowledge of this is important to any attempt we may make to understand the Australian conception of nature".

The Rainbow Serpent had many different roles depending on the tribe. Some of its roles related to: fertility in women and the land, close association

with medicine men and important ceremonies, protector of its people, land and the formation of features such as springs, rivers, lakes and lagoons, rain and flood events. The connection between the Rainbow Serpent and groundwater is of particular interest in this paper.

In western NSW there is deep connection between the Rainbow Serpent (*Wawi*) and both surface water and groundwater, connecting beneath and across landforms. In Rose et al. (2003), an account of this connection is given by Steve Meredith:

This country was made by the ancestors. *Wawi* the Rainbow Serpent came up through the springs, he came from Nakabo springs, Ngilyitri country. Wherever he travelled he left ochre to show where he had been. The springs were entry and exit points. He came out of the earth, travelled along its surface, and then back into the earth. *Wawi* travels and is still there. We know he's still there.

Rose et al. (2003) further explain that in Ngiyampaa country the *Wawi* came from the east, travelling underground, coming up in a spring in the Manara Range (near Ivanhoe, NSW) where he had a fight with Robin Red Breast, which he lost and is now stuck in the rockhole. *Wawi* rises out and returns as the water in the rock hole rises and falls.

Tindale (1974) mentions how Walmadjari men in times of drought (*lalba*), go to a big permanent waterhole, remove excess sand or soil from the hole and add to the height of the walls, making the hole deeper. Then they shout out very loudly with special cries to tell the *Wanambi* or giant carpet snake that they need water and to come and fill the well for them. It is believed that the giant hidden snake yields the seeping water to the Walmadjari. From a hydrogeological interpretation, these Walmadjari men have dug into the saturated zone, thus allowing groundwater to seep into the hole.

The association between the giant carpet snake and permanent waterholes in desert regions is so close that, for instance, when a waterhole dries up it is believed the snake has died. Under tribal lore, if a person mistreats a waterhole that affects the snake, punishment is inevitable.

These lores and customs would have been a survival trait, specifically to protect the water quality

and quantity of waterholes. Zaar et al. (2002) provide examples: "... at some waterholes, in order to avoid contaminating the water, people are not allowed to put their hands in to scoop water out", so for hygiene purposes they drink by sipping the water without using their hands. A Traditional Owner – Tony Djakanawuy – talks of protecting a stream called Darrangay in Arnhem Land: "... upper pools were used for drinking and the downstream pool was used for swimming" (Zaar et al., 2002). Another example is discouraging children from playing in springs by stating a serpent or bad spirit will take them. These were simple ways to protect water quality.

The Rainbow Serpent is also culturally significant for the Baakantji people. Martin (2003) mentions a story that links a waterhole at Union Bend, Wilcannia (western NSW) with the *Ngatyi* or Rainbow Serpent. This site is part of a series of *Ngatyi* waterholes along the Darling River that are important to the Baakantji people.

Yu & Yu (2000) record how the Karajarri people (south-west of the Kimberley region, Western Australia) describe the Dreaming species: *Pulany* or water snakes and serpents which reside in or have created permanent water sources – *jila* (spring) or *pajalpi* (spring country). The presence of *Pulany* in a spring is often indicated by the *panyjin* reed (species unknown), and is a warning sign for children not to swim there. *Panyjin* reeds are said to be the whiskers of the *Pulany*. Strangers should not approach a *jila* without the presence of a countryman for that area, who will call out to warn the *Pulany* of their presence and state their relationship.

Groundwater Stories from the Dreamtime

My thesis records many stories that indicate the linkages between surface water, groundwater, lakes and rivers, cave systems, natural springs, thermal springs, rain events recharging the aquifers, and how in drought excess discharge allowed cultural heroes to move with water-table fluctuations. Here I have selected three Dreamtime stories about springs:

The Freshwater Springs at Raymanggirr

At Raymanggirr, a place on the northern coast of Arnhem Land somewhere near Lake Evell, the grandchildren of the old frill-necked lizard man were collecting honey. The frill-necked

lizard heard, and stopped and listened. "Aha! My grandchildren are collecting honey!" He heard them chopping the tree to collect the wild sugarbag. "That tree's going to fall down," he said, and so he ran to a rocky point in the sea and looked up at the land. He named that point Mayawalpalnga and then he ran on.

Then his grandchildren called to him: "Come here, you dear old thing! You can have the top part of the honey in the tree, we'll have the bottom part." And they ate the honey. The lizard was eating his over there when he got something stuck in his throat. When we cook the frill-necked lizard, we still find these splinters in his throat. And he ran off into the bush calling, "A bit of the tree has stuck in my throat." He ran down to the edge of the sea water, into where the lily roots are. And here where the water runs into the sea, it is fresh. That old lizard man showed us where it is, and we can drink it today.

This is how the people collect it. They go down with a pannikin. When the tide's not full you can just collect it in a pannikin. But when the tide comes in and the spring is submerged, the water is collected in the mouth, sucked up into the mouth, and spat into a paperbark cup or pannikin. It's collected in the mouth. It is held in the mouth, carried over, and spat, and more is collected, and spat out, and more, and spat. Then it's carried to the camp and given to the people. "Sorry, not much water! The tide's in. Too much salt water. We'll go back later when we can dip for water properly."

If anyone objects to the spit, you can get a long hollow piece of wood like a bent didgeridu and the water will flow into it until it fills up. And another one, until that one fills up. Then carefully lift out the wood, and carry it to the camp and put it down. The water just flows by itself. When the tide's out, it runs down the beach. At Raymanggirr (Manuwa, Milingimbi, in Isaacs, 1980).

The Formation of Spring Waterholes in the Flinders Ranges

A family was walking from Curnamona to Barratta Springs. Because of dry weather there was no water and being so hot the family got thirsty. The old man told his family to walk

along slowly toward Mount Victor while he went to Barratta Springs to wait for a kangaroo. He wanted to kill it and make a waterbag out of its skin to carry the water back to his family, which by this time would have been somewhere near Mount Victor.

At Barratta Springs, the old man waited for the kangaroo which was coming from Murnpeowie. As the kangaroo hopped along, the old man could hear the thumping sounds. The next stop the kangaroo made was at Pepegooona Spring, near Wootana Station. From there, the kangaroo hopped on to Nurowi Springs and made water. There was no water there, so he hopped onto Emu Bore, which is now an artesian bore. From Emu Bore, the kangaroo hopped on to Limestone. He couldn't find any water there so he had a rest. The white ground at Limestone represents where the kangaroo rested. He then hopped to Glenwarrick Springs and from there he started hopping towards Barratta, but on his way he saw a big snake which now represents Tooths Knob Hill on the Martin's Well property. He got frightened so he went around toward Kemps Dam. There was no water there, so the only place he had to go was to Barratta Springs.

When he got there, the old man killed him, got his skin and made it into a waterbag, filled it up with water and headed off towards Mount Victor to find his family. When he got to Mount Victor he saw his family lying on the ground. They were dying, so he quickly poured water on them. As he poured the water on them it spread and formed a swamp or a kind of a lake. When he saw them moving and starting to come around, he jumped into the middle of the water and he saw his family turning into ducks. He disappeared into the sky and formed the Morning Star.

His family couldn't find him, so they looked into the sky and yelled out, "Look up there! That's our father looking down at us." That is how the springs were formed from Pepegooona Spring in the north down the eastern side of the Flinders Ranges to Barratta Springs (Eileen McKenzie, Flinders Ranges, in Isaacs, 1980).

Arkaroo's Dreamtime Journey

Back in the Aboriginal Dreamtime, a giant serpent known as Arkaroo, who was living in the

main water pound in the high Gammon Ranges south-west of Arkaroola, slithered down to the plains to quench his thirst. Arkaroo descended upon Lakes Frome and Callabonna and drank them both dry. The water was saline, and he became bloated. He dragged his heavily laden body back up towards the Gammons, and in doing so, he carved out the deep sinuous gorge that is now known as Arkaroola. On his way back he stopped at several places for a rest, and while doing so he formed springs and waterholes along the way. There are a few permanent ones around there now.

He dragged himself up into the Gammon Ranges, where he now sleeps safely in a hide-away at the Yacki Waterhole. Restlessly he sleeps on with his belly full of water, and whenever he turns the rumbling in his stomach sends out great noises that can be heard from time to time to this day. The minor earthquakes and rumblings have been recorded.

One of the most important waterholes left around these parts by the Arkaroo is that of the Paralana Hot Springs. The Aboriginals of long ago found this spring very convenient. They used to use it for domestic purposes, as well as bathe in it. It is said to have cured minor aches and pains. This spring became hot when, back in the old time, two young warriors fought for the love of a young girl. The victor, after killing his opponent, plunged his fire stick murder weapon deep into the spring, thus making it hot. Since then the water emerges only little below boiling point (May Wilton, in Isaacs, 1980).

Aboriginal Survival and Groundwater

The ability of Aboriginal people to survive in the Australian landscape, especially in desert regions engulfing approximately 70% of the continent, is somewhat astonishing. These regions of Australia undergo extreme variations such as low rainfall with high evaporation and large temperature variations between night and day.

Their ability to survive such conditions would not have happened in an instant; it would have developed over many thousands of years, when each aspect of knowledge gained by elders or tribespeople would have been passed down to the next generation. This suggests a highly developed

and sophisticated culture of teaching and learning, and the will to survive using all resources available.

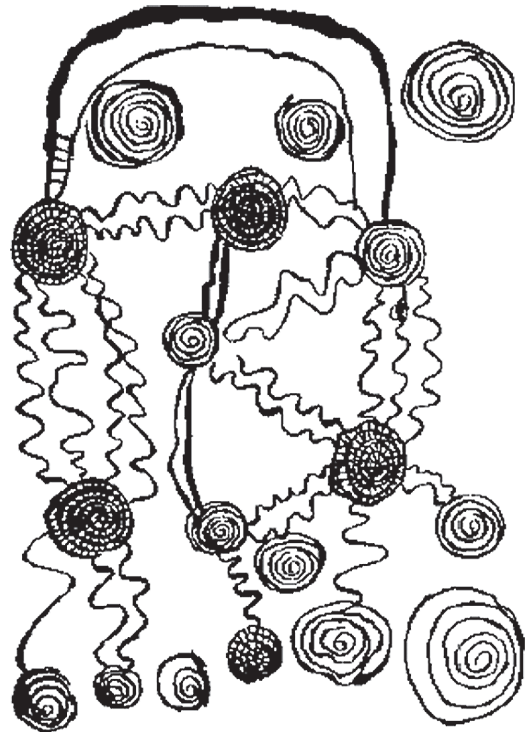
Aboriginal tribespeople developed a precise classification system for sites within their country. Their survival was dependent on this knowledge, and failure of this system would be fatal for the tribe. It was necessary to be precise when talking about waterholes, as life might depend on going to the right place at the right time (Lowe & Pike, 1990). A principal site for a tribe would have been a place to obtain water. Groundwater sources would have been significant, especially where waterbodies such as rivers or areas with high rainfall were not an option for water supply. Over thousands of years and to this day, Aboriginal people still know how to find or dig for water that has seeped up from subterranean sources.

The Walmadjari tribes' classification of water systems is given in Tindale (1974), and within this system groundwater sources are described as follows. They include waters trapped in deep sand which are classed as soak waters (*tju:mu*), and permanent waters classed as *tjila* and *tjaramara* which include natural springs (e.g. Joanna Springs in Mangala territory).

Another Aboriginal classification system for identifying groundwater sources through art is given in Figure 2 (Tindale, 1974). The drawing was done by Katabulka of the Ngadadjara tribe, who camped at Warupuju Soak in the Warburton Ranges, Western Australia. The original painting was four times as large and depicted in red and black. The painting depicts a map of water sources showing pools and soakages.

Across the top of it Tjurtirango the rainbow lies, and between it are two concentric spirals representing Kalkakutjara, the "heavenly breasts" [*kalka*] nipple and [*kutjara*] two, which gave rain that flows into [*kapi*] or waters. These are the balance of the concentric spirals. Five darker ones possess mythical [*koneia*] carpet snakes therefore are considered never-failing; the others are temporary waters. Down the middle runs a stream bed, dry except during rain. On it are marked three waters, of which the top one is Warupuju. Zigzag lines from water to water are the tracks or native roads of men wandering in search of food (Tindale, 1974).

Figure 2. Drawing by Katabulka of the Ngadadjara tribe in the Warburton Ranges in Western Australia. *Tjurtirango*, the rainbow, yields water to storage wells, and sand soaks symbolised by concentric spirals. Tracks made by men join the various waters (Source: Tindale, 1974). Available at <http://hdl.handle.net/1885/114913>



To find groundwater sources Aboriginal people would use all their available resources and natural indicators, which Tindale (1974) describes: "... in the arid zone, wild dingos preformed [*sic*] a service to Aboriginal people by locating and digging open water soakages." Following this, a digging stick would have been used to further dig out soakages to keep them clear of debris. Tindale (1974) also mentions the Nullarbor Plain where "a line of ants going down into a sinkhole in the limestone, can represent subterranean cave water" – karst groundwater.

Figure 3 is a reproduction of maps incised into spear-throwers, indicating the water resources of the Bindibu [=Pintupi] people of Great Sandy Desert, Western Australia. The narrative is given in Bayly (1999) and Brodie (2002) from the original description by Thomson (1962) at the end of his 1957 Bindibu [=Pintupi] Expedition when he

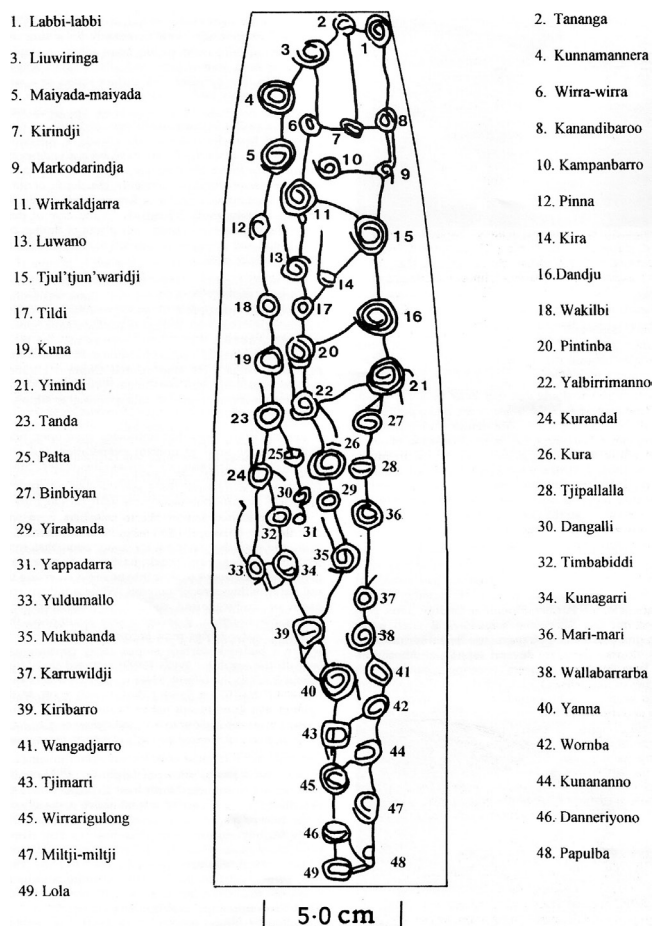
received the generous gift from Tjappanongo, who names and describes 49 water sources:

Just before we left, the old men recited to me names of more than fifty waters – wells, rock-holes and claypans – including those that I have described in this narrative; this, in an area that the early explorers believed to be almost waterless, and where all but a few were, in 1957, still unknown to the white man. And on the eve of our going, Tjappanongo produced spear throwers, on the backs [of] which were designs deeply incised, more or less geometric in form. Sometimes with a stick, or with his finger, he would point to each well or rock hole in turn and

recite its name, waiting for me to repeat it after him. Each time, the group of old men listened intently and grunted in approval – “Eh!” – or repeated the name again and listened once more. This process continued with the name of each water until they were satisfied with my pronunciation, when they would pass on to the next.

I realized that here was the most important discovery of the expedition – that what Tjappanongo and the old men had shown me was really a map, highly conventionalized, like the work on a “message” or “letter” stick of the Aborigines, of the waters of the vast terrain over which the Bindibu hunted.

Figure 3. A highly conventionalised map of the Western Australian water resources of the Bindibu [= Pintupi], as carved into the back of a spear-thrower (Source: Redrawn from a photograph in Thomson (1962) by Bayly (1999)). Available at [https://www.rswa.org.au/publications/Journal/82\(1\)/82\(1\)bayly.pdf](https://www.rswa.org.au/publications/Journal/82(1)/82(1)bayly.pdf)



Numerous natural indicators would have guided Aboriginal people to groundwater. Some are mentioned above, but further indicators are extensively described in Kavanagh (1984), including the terrain, birdlife, vegetation, animals and insects. Considering these indicators, Aboriginal people then had to access the aquifer for the precious water. A prime example of Aboriginal ingenuity is given in Burnum Burnum (1988), where the activities of the Central Lakes people, situated east of the Western Desert, had constructed tunnel-reservoirs to access underground water in the same way settlers later tapped artesian bores. Bandler (2000) mentions the structures and various formations, some natural, others man-made, which show a high degree of expertise and knowledge in assessing and preserving precious groundwater. The works were an integral aspect of Aboriginal culture, carried out with simple tools, as pottery and metals had not been introduced.

Aboriginal Art and Groundwater

Aboriginal art has played a significant role in classifying, representing and describing significant groundwater sites for Aboriginal tribes, for knowledge of water sources is so important to a tribe's survival. Aboriginal art was not only painted on canvases or linen as modern society now demands, but earlier Aboriginal people used many media, such as the body for ceremonies, rock shelters and platforms, ground designs (sand drawings and

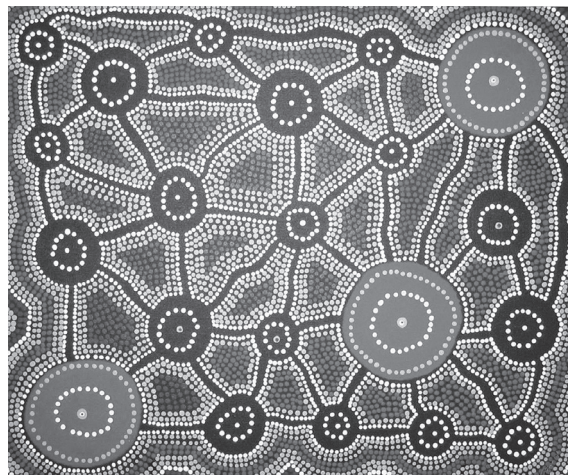
ground mosaics, also used for ceremonies), implements or artefacts, ceremonial poles and the bark off a tree.

Aboriginal art, especially that originating from desert regions of Australia and in the dot art form such as the Warlpiri and Pintupi Language Groups (north-west of Alice Springs, Central Desert of Australia), will constantly make reference to and represent groundwater sources such as soakages and springs. Some good examples of desert art indicating groundwater sources (springs), along with explanations, are given in Stokes (1993).

Aboriginal art uses traditional symbols that can be read in many ways. Because of this, even the secret/sacred parts of a story can be painted but still protected, for the artist is the only person who fully understands the meaning (Stokes, 1993).

Three Waters, painted by the author (Figure 4), is a personal story that I dreamt about and consequently painted onto canvas. The story was later confirmed by Kamilaroi elders as representing three soakages within the ancestral country. The painting represents three significant groundwater sources – soakages (large circles) which ancestors utilised throughout time. They are hand dug and consequently maintained en route or when the ancestors passed these sites. The small dark circles are camp sites along their travels, and the dark lines joining them indicate the travel routes taken between camps and the soakages.

Figure 4. A photo of a painting titled Three Waters (acrylic on canvas, 60 cm × 50 cm) painted by the Kamilaroi author of this paper (Source: B. Moggridge (2002) from personal collection).



Aboriginal art is described as the oldest continuous tradition of art known. Caruana (1987) states that there is no known fixed notion of traditional Aboriginal art, for it is not a static relic of a bygone era but a vital and pertinent expression of current human concerns. Through their art, Aboriginal people celebrate the ancestral mythologies which form the basis of their lives and cultures.

Aboriginal People and Groundwater Today

Aboriginal people would have chosen their place of settlement primarily for cultural, survival and social reasons. Permanent Aboriginal settlements have now replaced the former nomadic lifestyle, with many communities consisting of over 1000 people. Most of these settlements are in remote locations and rely on groundwater tapped by bores; supplying water to these communities is an ongoing challenge. Yuen et al. (2002) record that the challenge for remote Indigenous communities in Australia is to provide adequate supplies of potable and non-potable water to achieve health outcomes and meet cultural needs while minimising the economic, social and environmental costs.

A report by the Human Rights and Equal Opportunity Commission (2001) mentions that the number of remote Indigenous communities has grown over the last 20 years, largely due to the outstation movement. Further to this, Rowse (1999) explains that Indigenous communities as we know them today are a legacy of settlements around food rationing stations or mission and reserve establishments. Settled mixes of different family, tribal and skin groupings are new, and many of the social issues arise from the new types of community living that have no traditional basis for Aboriginal people.

From personal experience, the land where Aboriginal people are forced to settle in NSW is usually land of low value, highly degraded, unfamiliar country (not traditional lands) or adjacent to the local council's activities such as waste management centres and sewage treatment plants (Moggridge, 2003).

The availability of water is a crucial decision affecting an individual and their community's quality of life, as the life of the author's grandmother (Brenda Bengé née McGrady, 1918–2016) exemplifies. Nan was born in north-western NSW at Euraba Mission in 1918, and not long after her birth

the whole community was moved to Old Toomelah. Then, in 1938, the community was forced to move again to the current Toomelah settlement situated on the banks of the Macintyre River. These moves came about because of a lack of potable water (B. Bengé, pers. comm., 2002). The community currently uses a groundwater bore as their potable water supply.

Water availability and, in particular, permanency are often described as critical factors influencing the settlement patterns of Australian and other Indigenous peoples (Thorley, 2001). Water availability also pre-determines the distribution of plant and animal species. From an archaeological sense this is also reflected in the review of Thorley (2001) on water supplies and use in the Palmer River catchment, central Australia. This study displayed a close relationship between permanent waters and a wide variety of archaeological materials, whereas scatters near ephemeral waters were generally described as being less diverse. However, this pattern may vary around Australia, due to the significance of the water source to a particular tribe.

In desert regions, rainfall is unpredictable along with temperature extremes. The unpredictability of rainfall directs communities to groundwater for the main water supply (Gray-Gardner & Walker, 2002). A good example of groundwater's significance is given in Yu & Yu (1999), where a senior Karajarri man from the Great Sandy Desert, John Dudu, describes life without groundwater: "Water is the life for all of us. It's the main part. If that water go away, everything will die. That's the power of water. He connect with the land." Under their law, Karajarri are responsible for looking after their water; this would have been similar all through Aboriginal Australia.

Gray-Gardner & Walker (2002) indicate the total number of Aboriginal communities with a population of 100 or less using groundwater (Table 1). The discrete communities represented are concentrated in the more remote areas of the Kimberley and central desert regions and Arnhem Land.

Communities situated in remote areas using groundwater experience difficulties in securing access to bore water supplies of sufficient quality and quantity; also, the use of technology with regard to these supplies adds to the complexity of supplying infrastructure.

Table 1. Groundwater supplies for Indigenous communities with a population of 100 or less (Grey-Gardner & Walker, 2002).

State* or Territory	Number of communities	Number of communities on bore water	Per cent communities on bore water	Reported population in communities on bore water
Northern Territory	588	417	71%	8,226
New South Wales	35	6	15%	425
Tasmania	1			
Western Australia	235	182	78%	4,365
South Australia	90	47	52%	730
Queensland	109	22	20%	393
Total	1,058**	674		14,139

* Victoria and the Australian Capital Territory do not have communities with a population of 100 or less.

** Six Northern Territory communities did not state their type of water supply.

Within Aboriginal communities there is a constant fluctuation in residency numbers as movement between large centres and country may be influenced by a number of factors:

- Schooling.
- Family, e.g. 'sorry business' (funerals).
- Ceremonial or cultural reasons.
- Business or political meetings.
- Sporting events.
- Medical.
- Shopping.

These fluctuations in community numbers will place a strain on the community's infrastructure, i.e. water supply and wastewater disposal. It is integral for service providers to understand the cultural obligations of Aboriginal people (which differ from region to region). Planning the infrastructure for communities, depending on the cultural activity, can decrease a settlement to zero for up to six months, or the population may double for periods of time. Therefore, peak loads and demand will vary and the appropriate systems have to be in place to accommodate these patterns.

Studies of Groundwater Supplies and Services to Aboriginal Communities

Several studies have been carried out on Aboriginal communities in remote or desert country where community health and social issues are evident and service provisions are poor, especially regarding

potable water from groundwater sources. These studies have reviewed sustainability, social disadvantage, service provision, water quality and quantity, usage and management.

Review of the 1994 Water Report

In 1994, the Federal Race Discrimination Commissioner (RDC) released the Water Report, containing the findings of a comprehensive inquiry into the provision of water and sanitation services to Australia's remote Indigenous communities. A review of the 1994 Water Report was conducted by the Human Rights and Equal Opportunity Commission (2001) to assess developments over the past five years in 10 Aboriginal communities from around Australia between 1994 and 1999. The Centre for Appropriate Technology (CAT) in Alice Springs carried out the review for the Commission.

The 10 communities included: Punmu and Coonana from Western Australia, Yalata and Oak Valley/Maralinga from South Australia, Mpweringe-Arnapipe from the Northern Territory, Dareton and Tingha from New South Wales, Doomadgee from Queensland and two islands in the Torres Strait – Boigu Island and Coconut Island.

The Review documented significant improvements in water supply, with most communities accessing groundwater. However, communities still depend on an ongoing role of the state agency or regional service provider.

Western Water Study

The Western Water Study 1994–1997 was a ground-water study carried out by the Australian Geological Survey Organisation (AGSO) and partners (Central Land Council, the Northern Territory Department of Lands, Planning and Environment, and the Aboriginal and Torres Strait Islanders Commission – ATSIC). A summary of this study and its findings was given in Human Rights and Equal Opportunity Commission (2001).

The study area covered 68,000 square kilometres of central Australia. The main objectives of the study were to:

- improve access to groundwater information for Aboriginal people on their land;
- enhance environmental health;
- ensure equity in access to acceptable safe water, especially in remote and arid areas; and
- develop a rapid methodology that will provide these objectives.

Following evaluation, the study produced:

- a geographic information system (GIS) comprising: geological, hydrological and other pertinent data relating to water and the environment;
- a CD-ROM with all GIS information on water for the community to access and make decisions;
- a positive consultation and discussion process with the Aboriginal communities involved; and
- a 22-minute video recording produced by AGSO, which describes the study 'Water from Stone, Kapi mantanguru apurungu pakantja'.

Community Water Supplies in the Anangu Pitjantjatjara Lands

This study investigated the sustainability of nine major community water supplies in the Anangu Pitjantjatjara Lands, South Australia, between 1997 and 1999; the report was compiled by Dodds (2001). Groundwater derived from the 150 production bores is the only source of reticulated water in the Anangu Pitjantjatjara Lands, with many stakeholders involved in the

study. Results from the Fitzgerald et al. (1999) report are summarised here.

The groundwater supply is derived from fractured rock aquifers, which are discontinuous and limited in extent. Each community needs two bores with reasonable yields, and preferably three bores as a safeguard against bores drying up. Water quality is relatively good, with five of the nine satisfying Australian Drinking Water Guidelines and the remaining requiring treatment prior to consumption. Hardness of the Anangu Pitjantjatjara Lands groundwater is widespread.

The outcome of the project formed the scientific basis for the development of a regional water management strategy for the Anangu Pitjantjatjara Lands, with further consultation with communities as required. Eight recommendations listed below were produced by the study for relevant stakeholders to undertake (Fitzgerald et al., 1999):

1. The regional water management strategy should consider the scientific results reported here in conjunction with community aspirations and social, economic, political, and institutional factors. The responsibility for the strategy is likely to be with the new Arid Areas Catchment Water Management Board. The strategy should also consider likely water supply needs for economic development including pastoralism, mining or irrigation, also future needs for outstations.
2. We recommend that the bore monitoring program be continued for 10 years to provide the water use and water level data on which management of the borefields must be based and to determine the effect of recharge events of these groundwater systems.
3. The monitoring program needs some extension and changes: water level data is required for the borefields at the emerging communities of Watarru, Kanpi, and Nyapari; and some unpumped bores should now be monitored to obtain information on recharge without the complication of pumping (there are observation bores at Pukatja, Turkey Bore, and Iwantja available for this purpose). Creek flow should also be monitored with staff gauges in selected locations where groundwater recharge is dependent on creeks.

4. Water search leading to additional drilling is recommended for Mimili, Kalka, and Yunyarinyi (Kenmore Park) and is probably also needed at Watarru.
5. Water treatment (desalination) is recommended for community water supplies with unacceptable water quality such as those at Iwantja (Indulkana), Mimili, Amata and Kaltjiti.
6. Rainwater options should also be explored for the communities where groundwater quality is marginal to unacceptable. Rainwater can be a valuable source of drinking water, supplementing the supply of water from the bores. Stormwater harvesting, although probably less viable, should also be explored.
7. Our observations indicate the necessity of regular water quality monitoring especially for bacteria, and that appropriate field test kits are now available. The agency responsibilities for water quality monitoring need to be clarified.
8. Field trials of water conditioning of Amata carried out over 18 months appeared to successfully remove and prevent scale build-up in domestic installations. Use of this or similar units will reduce the high cost of maintenance and replacement of domestic hot water and other installations in these remote communities. There is a need for more extensive field trials of water treatment technologies for remote communities.

Summary and Conclusions

Groundwater is an integral component of the water cycle and classed as a finite resource – it is vast, travels through aquifers at a slow rate, and depends on local and/or regional geology. In Australia, it has become a favoured option for population centres as a potable water supply, but in some regions has been over-allocated, managed poorly and is now in danger of irreversible impacts. Careful forward planning by authorities is now required to avoid further groundwater depletion.

A vast number of Australian Aboriginal people and communities currently depend on groundwater but have a much longer history of use and connection (with timeframes of up to 65,000 years) compared to non-Aboriginal people. Their association with

groundwater started in the beginning – the Dreamtime – and is still strong today, when it is accessed through traditional as well as contemporary means. Aboriginal people used all available groundwater resources sustainably.

This long association with groundwater has resulted in Dreamtime stories related to sites being created and then passed on from one generation to another. The stories are re-enacted in ceremonies that capture the imagination of all, for educating and thus surviving. Many stories relate to individual groundwater sites as they are each significant.

Most traditional Dreamtime stories were lost following colonisation, assimilation and forced separation from country, but some survived and were passed on by tribespeople. Early settlers and anthropologists recorded some of these stories.

The Rainbow Serpent is a prominent and powerful cultural hero in many Dreamtime stories, and these stories relate closely to groundwater sites. The Rainbow Serpent had many different roles and names across the landscape, but sites associated with it are considered culturally significant.

The ability of Aboriginal people to survive in the Australian landscape, and especially in desert regions, is somewhat remarkable. This ability was dependent on a knowledge system that was precise; if not, it could have been fatal for a tribe. Knowing where to find water sites was a high priority, and Aboriginal people used all resources and indicators available to them for identification. These included: natural indicators (animals, insects), knowledge of the landscape and seasons, oral traditions (stories), and of course art, which has become popular in modern times.

Over many thousands of years, Aboriginal people have accumulated a comprehensive and astounding knowledge of groundwater. Part of this knowledge extends to sustainable management and use of groundwater. These factors need to be taken into consideration by governments – local, state or national – when planning decisions are to be made that may affect the quality or quantity of groundwater. If we want Australia to protect our groundwater resources for future generations, Aboriginal people must be involved in planning and decision processes.

Today, many permanent Aboriginal communities access groundwater for potable supply. This lifestyle is somewhat different from the pre-colonisation

nomadic lifestyle, and supplying adequate water to these communities is a challenge, as explained by Yuen et al. (2002) above.

The few studies conducted to understand Aboriginal people and their use of groundwater have been at a local or regional level, mainly investigating groundwater quality and quantity. However,

even fewer studies have looked at the close cultural relationship between Australian Aboriginal people and groundwater. This paper records the beginnings of my research on relationships, Dreamtime stories and cultural knowledge. My intention was, and still is, to inspire other Aboriginal people and researchers to take the subject matter further.

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